Download Yi-6B-200K locally

The goal is to download the pretrained Yi-6B-200K model from Hugging Face onto the user's local computer. This allows the user to access the model weights and configuration files before integrating the model into the project.

Step 1. Install Hugging Face Hub

Open a terminal (VS Code terminal, PowerShell, or Windows Command Prompt). Install the Hugging Face Hub Python package:

```
pip install -U huggingface_hub
```

This package provides the **hf** command-line interface (CLI), which allows users to download models directly from the Hugging Face Hub.

Verify that the installation was successful:

```
hf --version
```

If the command returns a version number, the CLI is ready to use.

Step 2. Download the model

The pretrained model is hosted in the following Hugging Face repository: **01-ai/Yi-6B-200K**.

Execute the following command:

```
hf download 01-ai/Yi-6B-200K --local-dir ./Yi-6B-200K
```

Explanation of the command:

- **hf download**: invokes the Hugging Face download utility.
- **01-ai/Yi-6B-200K**: specifies the model repository identifier.
- **--local-dir ./Yi-6B-200K**: specifies the destination directory on the local computer.

The CLI retrieves the model files from the Hugging Face repository and stores them in the specified directory.

> *Note: An internet connection is required. The download may take some time depending on the model size and network speed. Ensure sufficient disk space is available.*

Step 3. Verify the download

Once the download is complete, navigate to the directory where the command was executed. The following directory should have been created:

```
Yi-6B-200K/
```

To inspect its contents, execute:

```
dir Yi-6B-200K
```

On Linux or macOS, use:

```
ls -lh Yi-6B-200K
```

The directory should contain the model configuration, tokenizer files and pretrained weight files.

Expected structure (illustrative):

```
Yi-6B-200K/
|
|-- config.json
|-- generation_config.json
|-- tokenizer.json
```

```
|-- tokenizer_config.json
|-- special_tokens_map.json
|-- model.safetensors.index.json
|-- model-00001-of-0000N.safetensors
`-- model-00002-of-0000N.safetensors
```

What do these files represent?

- **config.json**: defines the model architecture and configuration parameters.
- **generation_config.json**: contains default text-generation settings.
- **tokenizer.json**: contains the tokenizer vocabulary and tokenization configuration.
- **tokenizer_config.json**: defines additional tokenizer settings.
- **model.safetensors**: contains the pretrained neural network weights.
- **model.safetensors.index.json**: maps model parameters to their corresponding weight shards when sharding is used.

The exact filenames and number of weight shards may vary depending on the repository revision and file format.

Step 4. Result

At this stage, the pretrained Yi-6B-200K model has been downloaded and stored locally on the user's computer.

Local model directory
`Yi-6B-200K/`
Download complete
The model weights and associated configuration files are available locally for subsequent use.